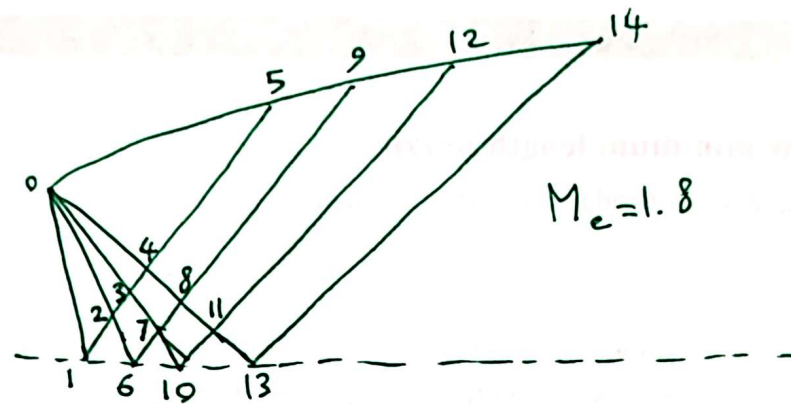


First, assume we have 4 expansion fans, we will have $\frac{n(n+3)}{2}$ (where $n=4$)
 = 14 points



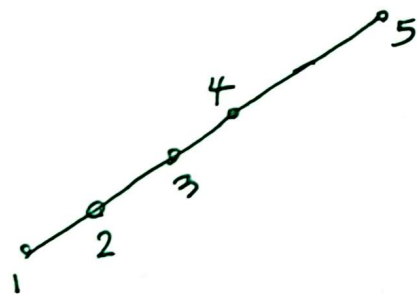
$$\nu(M_e) = 24.30252$$

$$\alpha_{\max, MLN} = \frac{24.30252}{2} = 12.1512^\circ$$

Assume the first wave turns the flow by 0.1512° (to start)

$$\nu_1 = \alpha_1 = 0.151 \xrightarrow{PM} M_1 = 1.0213 \quad \mu_1 = 76.28$$

The other 3 waves turn the flow by 4° (total 12.1512°)



$$C_{II} = \nu_1 - \alpha_1 = \nu_2 - \alpha_2 = \nu_3 - \alpha_3 = \nu_4 - \alpha_4 = \nu_5 - \alpha_5 = 0$$

$$\alpha_2 = 4.1512 = \nu_2 \rightarrow M_2 = 1.2 \rightarrow \mu_2 = 55.93$$

$$\alpha_3 = 8.1512 = \nu_3 \rightarrow M_3 = 1.33 \rightarrow \mu_3 = 48.32$$

$$\alpha_4 = 12.15 = \nu_4 \rightarrow M_4 = 1.45 \rightarrow \mu_4 = 43.29$$

wall turns as α_4 , so $\alpha_5 = \alpha_4 = 12.15$, $\alpha_{11} = \alpha_{12}$, $\alpha_8 = \alpha_9$, $\alpha_{13} = \alpha_{14} = 0$

①

$$\rightarrow v_5 = \alpha_5 = 12.151 \rightarrow M_5 = 1.4583 \rightarrow \mu_5 = 43.2931$$

$$\alpha_6 = \alpha_{10} = \alpha_{13} = 0$$

$$v_2 + \alpha_2 = v_6 + \alpha_6 = 8.3025 \Rightarrow v_6 = 8.30 \rightarrow M_6 = 1.34 \rightarrow \mu_6 = 4.0$$

$$v_3 + \alpha_3 = 16.30 = v_{10} + \alpha_{10} \rightarrow v_{10} = 16.3 \rightarrow M_{10} = 1.57, \mu_{10} = 3.0$$

$$v_4 + \alpha_4 = 24.30 = v_{13} + \alpha_{13} \rightarrow v_{13} = 24.30 \rightarrow M_{13} = 1.8 \rightarrow \mu_{13} = 33.74$$

Points 7, 8, 11 (need both C_I & C_{II})

$$C_I = \overbrace{v_3}^{8.15} + \overbrace{\alpha_3}^{8.15} = 16.3, \quad C_{II} = v_6 - \alpha_6 = 8.30$$

$$C_I = \overbrace{v_4}^{12.15} + \overbrace{\alpha_4}^{12.15} = 24.30, \quad C_{II} = \overbrace{v_7}^{12.3} - \overbrace{\alpha_7}^4 = 8.30$$

$$\begin{cases} v_p = \frac{C_I + C_{II}}{2} \\ \alpha_p = \frac{C_I - C_{II}}{2} \end{cases}$$

$\begin{matrix} 3 & C_I \\ & \searrow \\ & 7 \end{matrix} \quad v_7 = 12.30$
 $\begin{matrix} & C_{II} \\ & \nearrow \\ 6 & \end{matrix} \quad \alpha_7 = 4.00$

$\begin{matrix} 4 & C_I \\ & \searrow \\ & 8 \end{matrix} \Rightarrow v_8 = 16.30$
 $\begin{matrix} & C_{II} \\ & \nearrow \\ 7 & \end{matrix} \Rightarrow \alpha_8 = 8.00$

$$C_I = \overbrace{v_8}^{16.3} + \overbrace{\alpha_8}^8 = 24.30, \quad C_{II} = \overbrace{v_{10}}^{16.3} - \alpha_{10} = 16.30$$

$$v_{11} = 20.30, \quad \alpha_{11} = 4.00$$

$$\begin{matrix} 8 & C_I \\ & \searrow \\ & 11 \end{matrix}$$

$$\begin{matrix} & C_{II} \\ & \nearrow \\ 10 & \end{matrix}$$

At points 9, 12, 13 $\rightarrow$ we have boundary with known flow angle

$$\alpha_8 = \alpha_9 = 8 \quad , \quad c_{II} = \overbrace{v_8}^{16.3} - \overbrace{\alpha_8}^8 = 8.3025 = v_9 - \overbrace{\alpha_9}^8 \rightarrow v_9 = 16.3 \rightarrow M_9 = 1.57$$

$$\alpha_{11} = \alpha_{12} = 4 \rightarrow c_{II} = \overbrace{v_{11}}^{20.3} - \overbrace{\alpha_{11}}^4 = 16.3 = v_{12} - \overbrace{\alpha_{12}}^4 \rightarrow v_{12} = 20.30 \rightarrow M_{12} = 1.688$$

$$\alpha_{13} = \alpha_{14} = 0 \rightarrow c_{II} = \overbrace{v_{13}}^{24.3} - \overbrace{\alpha_{13}}^0 = 24.3 = v_{14} - \overbrace{\alpha_{14}}^0 \rightarrow v_{14} = 24.3 \rightarrow M_{14} = 1.8$$

$$\left. \frac{dy}{dx} \right|_I = \frac{y_P - x_A}{x_P - x_A} = \tan \left[\frac{(\alpha - \mu)_A + (\alpha - \mu)_P}{2} \right] = m_I$$

$$\left. \frac{dy}{dx} \right|_{II} = \frac{y_P - y_B}{x_P - x_B} = \tan \left[\frac{(\alpha - \mu)_B + (\alpha - \mu)_P}{2} \right] = m_{II}$$

$$\frac{A}{A^*} = 1.5398 \quad \left(\begin{array}{l} M=1.8 \\ \gamma=1.2 \end{array} \right) \quad \frac{A}{A^*} = \frac{A_{14}}{A_0} = \frac{v_{14}}{v_0} = \frac{1.5367}{1} = 1.5367$$

with our coars grid, the difference is 0.2%